MIL-W-80256A
25 August 1987
SUPERSEDING
MIL-W-80256
28 March 1980

MILITARY SPECIFICATION

WELDING MACHINES, ELECTRICAL RESISTANCE, SPOT, PRESS TYPE, AIR OPERATED, SINGLE PHASE, DIRECT CURRENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical resistance, spot welding machines of the single, phase, direct current (DC), air operated, press type with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2.1).

Class A - 18 inches throat depth.
Class B - 24 inches throat depth.
Class C - 36 inches throat depth.
Class D - 42 inches throat depth.
Class E - 48 inches throat depth.

Size 1 - 50 KVA at 50 percent duty cycle.
Size 2 - 75 KVA at 50 percent duty cycle.
Size 3 - 100 KVA at 50 percent duty cycle.
Size 4 - 150 KVA at 50 percent duty cycle.
Size 5 - 250 KVA at 50 percent duty cycle.

1.2.1 Throat depth. The welding machine throat depth is the clear unobstructed distance, measured in the welding plane, from the centerline of the welding electrodes to the first obstruction of the machine frame.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Machine size. The welding machine size is expressed in terms of kilovolt amperes (KVA) at 50 percent duty cycle of the welding transformer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-12332 - Welding, Resistance, Spot, Seam, and Projection for Fabricating Assemblies of Low-carbon Steel.

MIL-W-45210 - Welding, Resistance, Spot, Weldable Aluminum Alloys.

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

CFR 29, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise

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specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 5 - Resistance Welding Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, Washington, DC 20037-1580).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-1901).

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards.

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding machine shall be new and one of the manufacturer's current models capable of welding with electrodes mounted in a vertical plane in accordance with the requirements herein. The machine shall include all

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components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding machine. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the welding machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the welding machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the welding machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the welding machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 that are applicable to the machine. The welding machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard

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required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The welding machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The welding machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forging. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

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3.3.6 Threads. All threaded parts used on the welding machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose.

3.4 Components. The welding machine shall consist of not less than a frame and base, an enclosure, a head and ram assembly, a lower knee and horn, electrodes and electrode holders, an electrical system and controls, a pneumatic system, and a water cooling system.

3.4.1 Frame and base. The frame and base assembly of the welding machine shall be a steel weldment of box construction, ribbed to maintain original alignment and to minimize deflection. The base shall provide for floor mounting. Access to the interior shall be provided to permit inspection and servicing of the equipment installed within the frame. Doors and access panels shall be provided with locks and interlocks to prevent access by unauthorized personnel to live portions of the machine. The frame shall be designed to permit positioning of the lower knee and horn assembly.

3.4.2 Enclosure. The enclosure shall be constructed of corrosion resistant metal or metal treated to resist corrosion. Ventilation and cooling shall be adequate to prevent temperatures in excess of design requirements under normal operating conditions of rated power output.

3.4.3 Head and ram assembly. The head and ram assembly shall be designed to maintain original alignment with minimum deflection under load. The guiding mechanism for the ram shall be of the anti-friction type, provided with adjustment for wear, and shall be protected against dust and foreign matter. The ram movement shall be operated by an air cylinder that is actuated by a foot switch. The upper electrode shall be electrically connected to the secondary windings of the transformer by a silver plated, laminated flexible shunt. The ram stroke shall conform to the requirements specified in table I. The ram shall be cushioned for minimum shock operation.

Table I. Ram stroke, horn, and electrode holder capacities.

Machine size	Ram stroke (inches)	Horn diameter (inches)		Electrode holder diameter	Electrode holder taper number
		min	max		
1	3	2-1/4	2-3/4	1-1/4	RWMA #5 (#2 Morse)
2	3	2-1/4	2-3/4	1-1/4	RWMA #5 (#2 Morse)
3	3	2-3/4	3	1-1/2	RWMA #7 (#3 Morse)
4	4	3	3-1/2	1-1/2	RWMA #7 (#3 Morse)
5	4	3	3-1/2	1-1/2	RWMA #7 (#3 Morse)

NOTE: Requirements are not less than shown.

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3.4.4 Lower knee and horn. The lower knee shall be mounted on the frame of the machine to provide a vertical adjustment range of not less than 6 inches. A hydraulic jack shall be permanently mounted on the base of the machine to aid in the adjustment of the arm. The knee shall have a socket to permit motion of the horn and alignment of the lower electrode holder. The socket shall have a clamp that will not mar the horn. The horn shall be made of solid copper and have a diameter as specified in table I. The horn shall have clamps that will not mar the electrode holders. With maximum rated electrode force applied and with the lower arm and horn unsupported by any bracing means, the difference between the horizontal deflection of the upper electrode and the lower electrode and the vertical deflection of the lower holder shall not be more than the values specified in table II.

Table II. Deflection limits.

Machine size	Horizontal deflection difference between the upper and lower electrodes (inches)	Vertical deflection of the lower electrode holder (inches)
	All throat depths	All throat depths
1 and 2	.005	.040
3	.007	.040
4 and 5	.010	.040

3.4.5 Electrodes and holders. The electrodes and holders shall conform to the requirements of table I and RWMA Bulletin Number 16. Unless otherwise specified (see 6.2.1), one set of electrode holders 8 inches in length and one set of straight electrodes of RWMA group A, class 2 alloy, with a style F nose shall be supplied.

3.4.6 Electrical system. The electrical system of the welding machine shall include the transformer, rectifiers, timers, meters, circuit protection, disconnect switches, and all controls required for operation. The electrical system and components shall conform to the applicable requirements of NFPA 70. Each sub-panel shall be arranged so that it may be removed independently of any other sub-panel. All terminals and units shall be clearly identifiable from the machine electrical wiring diagram. Adequate ventilation of the control cabinets shall be provided. Where ventilation is provided by forced air, the air inlet shall be provided with replaceable filters.

3.4.6.1 Electric service requirements. Unless otherwise specified (see 6.2.1), the machine shall operate on a 220/440 single phase, 60-Hz, primary electric service. The machine shall be initially wired for operation on 440-volts.

3.4.6.2 Main disconnect switch. The machine shall have a main disconnect switch or a circuit breaker. The main disconnect or circuit breaker shall

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provide a safety interlock when access doors are opened to areas that present a hazard to the operator. Unless otherwise specified (see 6.2.1), a circuit breaker shall be furnished and shall be mounted on the machine panel.

3.4.6.3 Welding transformer. The welding transformer KVA rating and secondary current output shall be in accordance with table III for the machine size specified. The transformer shall be water cooled and thermostatically controlled to minimize condensation. A temperature responsive device shall be provided to prevent energization of the transformer if the temperature of the transformer exceeds design limits. All static connecting surfaces of the secondary circuit shall be silver plated. The transformer shall be insulated to conform to RWMA Bulletin Number 16 for class B, or better, insulation. The transformer shall have an 8 position tap switch in the primary. When specified (see 6.2.1), the transformer shall have a series/parallel switch in the primary circuit.

3.4.6.4 Secondary current rectifier assembly. The secondary current rectifier shall be of the silicon diode type using the single-phase, half wave or full wave circuit arrangement. The forward current rating and reverse peak voltage rating of the diodes shall be compatible to the size machine specified. The silicon diodes shall be mounted in a water cooled conductor. The conductors and diodes shall be electrically symmetrical. The diodes shall be arranged for ease of examination, installation, or removal. Surge suppression protection shall be provided with sufficient capacity to reduce any transient voltage peak which would cause failure of the diodes. The rectifier system shall have a protective control circuitry which will open the line circuit breaker upon excessive current and over temperature.

3.4.6.5 Solid state components. Solid state design shall be used throughout the welding machine for electronic components with only such relays as required to assure adequate safety and reliable operation to be used. Relays shall be the plug-in type where practical.

3.4.6.6 Electrical controls. The electrical controls shall consist of the welding power control, timing and sequence controls, current adjustment control, welding contactor, and auxiliary and accessory controls. Operator controls shall be located so as to be readily accessibly to the operator. All calibrations, phase shift controls, and similar characteristics shall be clearly marked in standard terms.

3.4.6.6.1 Operator control panel. The operator control panel shall have an "on-off" or "weld-no weld" switch, a "power on" switch and sequence lights, a current adjustment control, a weld timer selector, auxiliary and accessory control function switches, and any other controls or switches required to meet the requirements specified herein.

3.4.6.6.2 Welding power control unit. The welding machine shall be furnished with a synchronous, precision, combination welding power control unit conforming to, or equivalent to, NEMA ICS 5, type S2H. The control unit may be located in a separate sheet metal cabinet or mounted as an integral part of the machine.

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Table III. Machine output characteristics.

Machine size	Welding transformer power rating (KVA)	Spot weld throat depth(in.)	Electrode force applied 1/ (lbs)	Welding current 2/ (amps)	Welding two equal thickness					
					Mild steel		Stainless steel		Aluminum alloys	
					min.	max.	min.	max	min.	max.
1	50	18	2250	27,000	.016	.141	.020	.080	.016	.032
		24	1550		.016	.109	.020	.062	.016	.032
		36	1275		.016	.094	.020	.050	.016	.032
		42	1000		.016	.074	.020	.044	.016	.032
2	75	18	4000	40,000	.020	.250	.025	.141	.020	.062
		24	3500		.020	.200	.025	.125	.020	.062
		36	3000		.020	.187	.025	.109	.020	.062
		42	2250		.020	.141	.025	.080	.020	.062
		48	2250		.020	.141	.025	.080	.020	.062
3	100	18	7600	50,000	.030	.350	.030	.219	.025	.081
		24	6300		.030	.344	.030	.187	.025	.081
		36	5000		.030	.312	.030	.156	.025	.081
		42	4000		.030	.250	.030	.141	.025	.081
		48	3500		.030	.200	.030	.125	.025	.081
4	150	18	8000	60,000	.035	.380	.035	.230	.032	.102
		24	7600		.035	.375	.035	.219	.032	.102
		36	6300		.035	.344	.035	.187	.032	.102
		42	5000		.035	.312	.035	.156	.032	.102
		48	4500		.035	.250	.035	.150	.032	.102
5	250	18	9100	80,000	.040	.437	.040	.375	.040	.125
		24	9100		.040	.437	.040	.375	.040	.125
		36	9100		.040	.437	.040	.375	.040	.125
		42	7600		.040	.375	.040	.250	.040	.125
		48	7600		.040	.344	.040	.219	.040	.125

NOTE: Requirements are not less than those shown.
1/ The pneumatic system supplied with 80 psi air pressure.
2/ Maximum amperes with aluminum being welded.

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3.4.6.6.3 Timing and sequencing control. The control equipment shall provide operating control and adjustment of the welding variables to ensure that optimum values of force and current timing and sequencing can be repetitively attained to produce uniform spot welds in accordance with this specification. The welding machine shall be furnished with a sequence timer conforming to, or equivalent to, NEMA ICS 5, type 7B. The timer shall have at least the following functions: squeeze, weld, hold, and off. When specified (see 6.2.1), the controls shall be equipped with functions necessary to perform welds for aluminum alloys as specified in MIL-W-45210. When specified (see 6.2.1) additional auxiliary control functions shall be furnished. The controls shall have signal lights on the panel to indicate the operating sequence of the welding functions and to be used as an aid in trouble shooting when a malfunction occurs. The various timing and sequencing function shall be grouped on a plug-in chassis with provisions for the removal or addition of extra functions and to allow the maximum ease of maintenance or replacement.

3.4.6.6.4 Current adjustment control. Continuously variable adjustment of the welding current, from the upper to the lower limits of the machine capacity, shall be provided. The control shall be graduated in increments of 1 from 0 to not less than 99 or from 10 percent to not less than 99 percent for convenience in identifying settings. The means of current adjustment provided shall be such that, at the maximum setting, the machine's maximum welding current produced is not less than the value specified in table III for the machine size specified.

3.4.6.6.5 Welding contactor. Unless otherwise specified (see 6.2.1), the welding contactor shall be the silicon controlled rectifier (SCR) type. The SCR contactor shall be mounted in a heat sink and shall be water cooled and thermostatically protected. Means shall be provided to prevent welding if the water is shut off or if the supply is restricted. The SCR contactor and the firing circuit element shall be protected from excessive current demands. A light shall be provided to indicate when the protective device has been energized.

3.4.6.6.6 Automatic voltage compensation. When specified (see 6.2.1), the welding machine shall be provided with an automatic voltage compensator (AVC) to minimize the effects of line voltage variations. The AVC shall maintain the welding current within ±2 percent for a line voltage variation of ±10 percent. Full compensation shall occur within 1-1/2 cycle for any change in line voltage.

3.4.6.6.7 Control circuit voltage. Control equipment shall be capable of operating on control circuit voltage which varies as much as ±10 percent from the voltage for which the equipment is adjusted. All control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 24 volts.

3.4.6.6.8 Accessory controls. The accessory controls shall consist of at least a foot switch, a tip dress control, an electrode force interlock circuit, a weld "on/off" or "weld/no weld" switch, and a repeat "on/off" or "repeat/non-repeat" switch.

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3.4.6.6.8.1 Foot switch. The foot switch shall control the upper electrode assembly and its associated pneumatic system, and initiate the welding cycle. Unless otherwise specified (see 6.2.1), a two-stage foot switch shall be supplied which shall initiate the electrode force in the first stage, and initiate the welding sequence in the second stage. The foot switch shall be guarded to prevent accidental operation of the machine.

3.4.6.6.8.2 Tip dress control. The tip dress control shall operate to lower the upper electrode under a reduced force for cleaning and aligning the electrodes. When applicable, the control may be activated with the first stage of the two-stage foot switch.

3.4.6.6.8.3 Electrode force interlock circuit. The electrode force interlock circuit shall prevent the upper electrode from being raised during the welding cycle.

3.4.6.6.8.4 Weld on/off or weld/no weld switch. A "weld/no weld" switch shall be provided. In the "off" or "no weld" position, the control sequence will actuate except that no welding current shall flow.

3.4.6.6.8.5 Repeat on/off or repeat/non-repeat switch. A "repeat on/off" switch shall be provided. In the off or non-repeat position only one welding cycle will result with the actuation of the switch. In the "on" or "repeat" position, the welding cycle will repeat automatically after a controlled time interval (off time) as long as the switch remains closed.

3.4.6.6.9 Optional accessory controls. Optional accessory control may consist of the following.

3.4.6.6.9.1 Contact pressure switch. When specified (see 6.2.1), the machine shall have a contact pressure switch that shall prevent the initiation of the welding cycle until the predetermined air pressure in the pneumatic head is equal to the contact pressure setting.

3.4.6.6.9.2 Electrode forge force control. When specified (see 6.2.1), the machine shall have an electrode forge force control that shall permit selection of a constant high, constant low, or a variable force sequence at the electrodes for welding.

3.4.7 Pneumatic system. The pneumatic system shall conform to the requirements of RWMA Bulletin Number 16 where applicable, and not in conflict with the requirements of this specification.

3.4.7.1 Pneumatic system operation. Unless otherwise specified (see 6.2.1), the pneumatic system air cylinder shall be a double action, adjustable stroke type, and shall be designed to operate with 80 pounds per square inch (psi) input air service. The inlet connection shall be marked for the pressure requirements. When the adjustable retractable stroke system is specified, the adjustable total stroke shall be not less than 4 inches. The ram and upper electrode pneumatic system shall include valving, piping, regulators, gauges, lubricators, moisture drain provisions, wiring, switches,

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and all related equipment necessary to permit use of any of the electrode forces. Air cushions or air speed valves shall be provided to ensure minimum mechanical or pneumatic shock operation. The system shall be controlled by a solenoid valve actuated by a foot switch.

3.4.7.2 Air gauges. Air gauges shall have a face diameter of not less than 4-1/2 inches, an accuracy of ±0.5 percent of the maximum gauge reading, and when not specified, calibrated in psi. When specified (see 6.2.1), the gauge shall be calibrated in pounds of applied electrode force. Means of shock protection shall be provided to prevent gauge damage. The gauges shall be mounted on the front or side of the machine and be visible to the operator.

3.4.7.3 Air regulators. The regulators shall be of the self-bleeding type. Regulator and solenoid valves shall be installed with unions. The regulators shall maintain the air pressure at ±1/2 psi within a 60 to 100 psi range as long as the supply pressure is 2 psi higher than the preset regulator pressure. The air system for the upper electrode shall produce the required electrode force to perform the welds specified herein. When specified (see 3.4.6.6.9.2), the rise of the electrode force, from its weld force level to its forge force level, shall be 75 percent accomplished in no more than 20 milliseconds.

3.4.8 Water cooling system. The machine shall have a water circulation system that will prevent the temperature rise during operation from exceeding the limits specified in RWMA Bulletin Number 16. The system shall operate from a shop tap water supply and shall have a single inlet connection into a manifold. The inlet connection shall be marked for pressure and flow requirements. The system shall have parallel flows from the manifold to the welding transformer, to the secondary winding and circuit conductors, to the primary welding current contactors, and to the upper and lower electrodes. The piping in the system shall be corrosion-resistant to tap water service. The system shall have an inlet strainer and a closed manifold drain system with flow indicators. The water flow lines to the electrodes shall have solenoid valves to prevent water flow while changing the electrodes. When specified (see 6.2.1), the system shall have water saving devices.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the welding machine shall be not less than the requirements in tables I, II, and III for the machine size specified.

3.6 Performance requirements. When connected to the specified source of electric, air, and water service, the welding machine shall conform to the requirements of this specification and shall produce spot welds conforming to the requirements of MIL-W-12332 and, when specified, the requirements of MIL-W-45210 for aluminum alloys.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all standard equipment normally furnished with the manufacturer's standard commercial welding machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

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3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, stenciling, silk screening, or by any other method suitable for the use intended. Marking shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be securely attached to each welding machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A corrosion-resistant metal nameplate shall be securely attached to each welding machine. The nameplate shall contain the information listed below, and that information required in RWMA Bulletin Number 16 for nameplate data. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Date of manufacture
Contract or Order Number
National Stock Number or Plant Equipment Code

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the welding machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The welding machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The welding machine shall be operated at no load

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for not less than 15 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance test. The welding machine shall be tested for welding capability in accordance with paragraph 3.6. Thirty welds shall be made using two thicknesses of the maximum mild steel specified in table III for the machine size specified. The welds shall be examined and tested in accordance with MIL-W-12332. When aluminum alloys welding capability is specified, 30 consecutive welds shall be made using two thicknesses of the maximum aluminum alloys specified in table III for the machine size specified. The welds shall be examined and tested in accordance with MIL-W-45210

4.7.3 Automatic voltage compensation test. When the AVC is specified (see 3.4.6.6.6), the performance of the AVC shall be measured by means of a calibrated recording instrument. With the current adjustment set at 50 percent and then at 90 percent of maximum, the secondary current and primary voltage shall be recorded as ±10 percent step changes are introduced into the primary voltage. The compensation and response time may be tested by causing the voltage change to occur only on the control of the compensator. The 10 percent increase or decrease in voltage shall cause, respectively, a decrease or increase of 10 percent in the secondary current within ±2 percent. The secondary current compensation response time shall be within 1-1/2 cycle as specified in 3.4.6.6.6.

4.7.4 Dielectric test. The machine shall be subjected to the dielectric test described in RWMA Bulletin Number 16.

4.7.5 Electrode force test. The maximum electrode force of the machine shall be measured, using an electrode force gauge. The measured maximum shall be not less than the values specified in table III.

4.7.6 Forge force rise time. When the electrode force control is specified (see 3.4.6.6.9.2), the machine sequence shall be set in the variable force selection, and with the machine controls adjusted to the proper values for welding two pieces of the maximum thickness of aluminum alloys as specified in table III, the rise time of the forge force and the time relationship of the weld current and the forge force shall be measured by means of a calibrated recording instrument. The incremental rise of the electrode force in the variable pressure mode from its weld force level to its forge force level shall be 75 percent accomplished in not more than the time specified in 3.4.7.3.

4.7.7 Electrode deflection test. The machine shall be tested to determine the vertical and horizontal deflection of the upper and lower electrodes. The electrodes shall be positioned at the nominal throat depth location, and the pneumatic system adjusted to provide the machine's maximum electrode force value specified in table III. Without the lower arm or horn being supported by a brace or any other device that can serve as a brace, the electrodes shall be brought to bear upon a plate and roller assembly as illustrated in figure 1. The diameter of the rollers and the thickness of the plates shall be not less than one-half the electrodes diameter and preferably not more than the value. Under these conditions, the difference between the upper electrode and the lower electrode horizontal deflection and the vertical deflection of the lower electrode holder shall be not more than that specified in table II.

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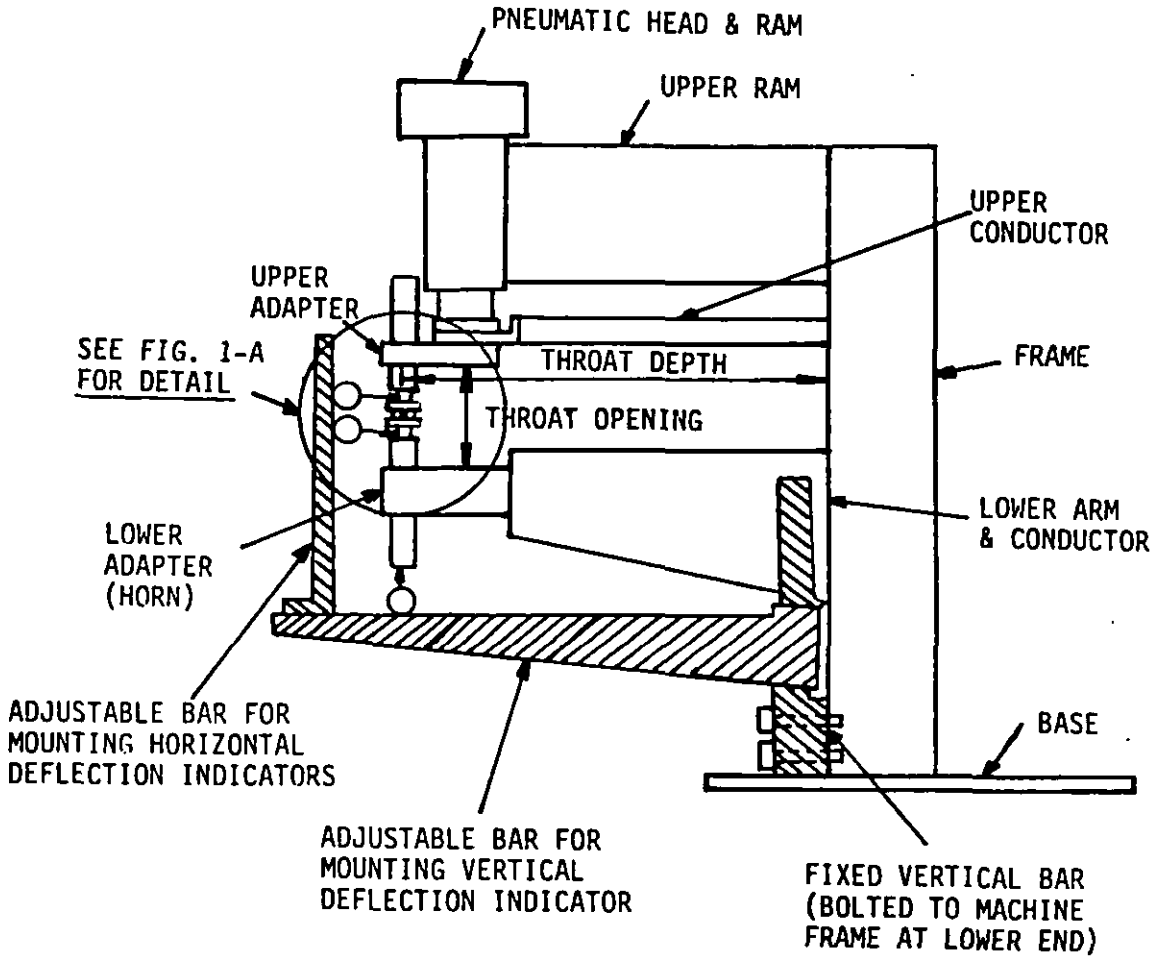


FIGURE 1. TEST SET-UP FOR ELECTRODE HORIZONTAL & VERTICAL DEFLECTION MEASUREMENTS.

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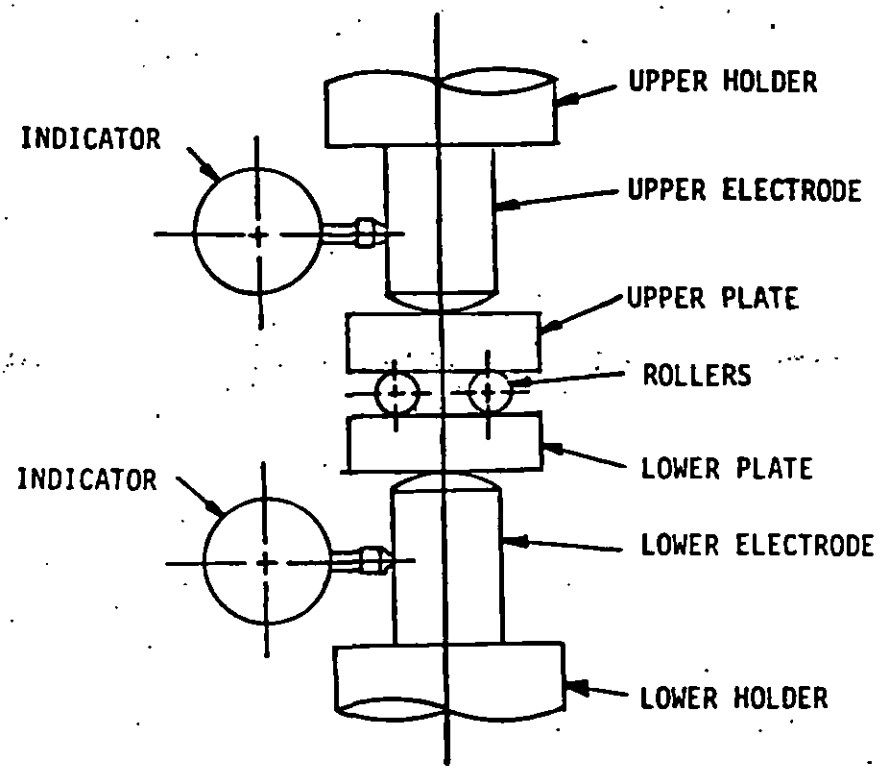


FIGURE 1A. DETAIL OF DIAL INDICATOR SET-UP FOR MEASURING ELECTRODES HORIZONTAL DEFLECTION

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4.7.8 Secondary current test. The maximum secondary welding current shall be determined and it shall comply with the values specified in table III.

4.7.9 Rated capacity test. The welding machine shall be adjusted for welding the maximum thicknesses of material specified in table III for the machine size specified and operated for 1 hour of simulated welds. The temperature limits specified in RWMA Bulletin Number 16 shall not be exceeded during the test. There shall be no evidence of overheating, arcing, or malfunction of the welding machine.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The welding machines covered by this specification are intended for spot welding metals which are not appreciably hardened by quenching from elevated temperature and under production conditions in manufacturing plants, shipyards, and repair shops, and where it is desirable to use single phase (DC) equipment for welding. An example of the use is spot welding of weldable aluminum alloys intended for storage boxes, vehicle fenders, air ducts, van skin and other similar items.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size, required (see 1.2).

c. First article, if required (see 3.1).

d. If welding machine design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), stated required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

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h. Painting, if different (see 3.3.5).

i. Electrodes and holders, if different (see 3.4.5).

j. Electrical system and input power, if different (see 3.4.6.1).

k. Circuit breaker and its mounting, if different (see 3.4.6.2).

l. Series/parallel switch, if required (see 3.4.6.3).

m. Control functions for welding aluminum alloys, if required (see 3.4.6.6.3).

n. Auxiliary control functions, if required (see 3.4.6.6.3).

o. Welding contactor, if different (see 3.4.6.6.5).

p. Automatic voltage compensator, if required (see 3.4.6.6.6).

q. Two stage foot switch, if different (see 3.4.6.6.8.1).

r. Contact pressure switch, if required (see 3.4.6.6.9.1).

s. Electrode forge force control, if required (see 3.4.6.6.9.2).

t. Pneumatic system air cylinder, if different (see 3.4.7.1).

u. Air gauge calibrated in pounds of force, if required (see 3.4.7.2).

v. Water saving devices, if required (see 3.4.8).

w. Size and capacity, if different (see 3.5).

x. Additional performance features, if required (see 3.6.1).

y. Standard equipment, if different (see 3.7).

z. Specify and fully describe optional equipment, if required (see 3.8).

aa. Specify repair parts, if required (see 3.9).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (see 4.5).

dd. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing required (see 5.1).

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6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The welding machine classification changes from the previous issue of this specification are as follows:

MIL-W-80256	MIL-W-80256A
18 inches throat depth	Class A
24 inches throat depth	Class B
36 inches throat depth	Class C
42 inches throat depth	Class D
48 inches throat depth	Class E

6.5 Subject term (key word) listing.

Automatic voltage compensator
Electrode, copper
Electrode deflection
Electrode forge force
Electrode holder
Resistance welding standard
Sequence timer
Synchronous welding power control
Throat depth
Tip dress control
Welder, resistance
Welder, spot
Welding contactor
Welding electrodes
Welding machine, resistance spot, press type
Welding transformer

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Air Force - 99

Preparing activity:
DLA - IP

(Project 3432-0137)

Review activities:
Navy - AS, SH
Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80256A	Welding Machines, Electrical Resistance, Spot, Press Type, Air Operated, Single Phase, Direct Current

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
DIPEC-SSG	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)* The Defense Logistics Agency Defense Industrial Plant Equipment Center 2163 Airways Blvd. Memphis, TN	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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